

Post-Earnings-Announcement Drift in the IBES Sample

A decile-sorted event study of analyst-based earnings surprise, 1996–2013

Data: WRDS I/B/E/S consensus and actuals, CRSP daily returns, Compustat classification · 182,712 analyst-sourced earnings events across 6,766 tickers (see Section 1 for full universe and coverage scope)

This report tests whether the post-earnings-announcement drift (PEAD) anomaly is present in this project's cleaned IBES/CRSP dataset. PEAD is the well-documented finding that stock prices continue to move in the direction of an earnings surprise for weeks after the announcement, rather than adjusting immediately — a pattern that is difficult to reconcile with market efficiency. The test here is the standard one in the literature: sort earnings events into ten groups (deciles) by the size of their standardized unexpected earnings (SUE), then track each decile's average market-adjusted return over the following 1 to 60 trading days. If PEAD is present, decile 10 (the most positive surprises) should drift up, decile 1 (the most negative surprises) should drift down, and the ordering in between should be roughly monotonic.

Result: yes, cleanly. Across every horizon tested (1, 5, 10, 20, 40, and 60 trading days), the ten deciles are monotonically or near-monotonically ordered, the spread between decile 10 and decile 1 is positive and grows with the horizon, and the spread remains statistically significant after clustering standard errors by announcement quarter (the project's own stated remedy for the one caveat — earnings-season clustering — that the underlying pipeline documentation flags as unaddressed). A size-and-book-to-market risk adjustment (Section 4b) shows the drift is not simply compensation for holding riskier stocks: the adjusted spread is, if anything, larger than the raw one. The effect is not industry-specific: a formal test finds no significant difference in drift magnitude across the twelve Fama-French industry sectors. It is, however, strongly size-related: small-cap stocks drift roughly four times more than large-cap stocks, consistent with the standard limits-to-arbitrage explanation for why this anomaly persists. Section 6 turns the signal into three tradeable portfolio constructions and backtests them net of transaction costs and liquidity limits: all three are profitable (Sharpe 0.49-0.56 over 1996-2013), with a rank-weighted construction that trades the full cross-section outperforming the simpler extreme-decile long/short.

1. Data and sample construction

Universe scope. Every number in this report is drawn from this project's existing WRDS pull, which was already restricted at the pull stage to the **OptionMetrics-linked universe**: 8,358 secids (8,115 distinct CRSP permnos) — every US common stock that had listed options on it at some point in this project's window, per `data/metadata/optionmetrics_universe.parquet` and the `universe_fingerprint` recorded in the CRSP pull manifest. This is a broad cross-section (it spans all five size quintiles roughly evenly, from micro/small caps to mega caps — see Section 5b), not a narrow index like the S&P 500, but it is **not the full CRSP/IBES universe**: a small-cap or micro-cap stock that never had listed options at any point is not in this data at all, regardless of whether it had earnings announcements. Expanding past this universe would require a fresh WRDS pull, which this session cannot perform (no WRDS credentials are available here) — the raw `data/raw_wrds/` pull that this project's own pipeline is built on already happened outside this analysis, at an earlier stage not reproducible from this environment. All results below should be read as applying to the optionable-stock universe specifically.

Ticker coverage within that universe. This report does not use all 8,115 permnos either. Of those, 7,222 have at least one earnings event with both a usable IBES match and a resolved day-0 trading session in `equity_events_1996_2013.parquet` (the remaining 893 have no usable earnings-announcement event of any kind in this data, unrelated to the choice below). Of those 7,222, this report is further restricted to the 6,766 tickers (93.7% of the 7,222; 83.4% of the full 8,115) that have **at least one analyst-sourced** (`sue_source == 'analyst'`) **event**. The remaining **456 tickers are entirely absent from this report** — every one of their earnings events used only the seasonal (year-over-year) surprise measure or had no usable surprise measure at all, and were dropped along with that filter, not sampled out. This was a deliberate scope choice (Section 1 already explains why: pooling the analyst and seasonal measures without adjustment mixes two differently-scaled surprise constructions), not a coverage gap that appeared by accident. Recovering those 456 tickers would mean re-running the sort on the pooled, rank-normalized surprise measure (`sue_rank_pct`, as the project's own existing quintile analysis already does) instead of the analyst-only measure used here.

This analysis builds directly on data already cleaned and validated earlier in this project (see `data/earnings/FINDINGS.md`, `data/events/FINDINGS.md`, and `data/results/FINDINGS.md` for the full derivation, all point-in-time and split-adjustment fixes, and their measured effects). In summary, the underlying event table already:

- restricts every earnings-surprise measure to information that was strictly public before the announcement (a hard `statpers < anndats` filter, i.e. no look-ahead bias);
- corrects both the analyst estimate and the reported actual for stock splits between the consensus snapshot and the announcement (three known extreme-outlier events, e.g. FEMP going from -104.9 to +0.87, were pure split artifacts before this fix);
- computes forward returns from the first trading day at/after the announcement (“day 0”), market-adjusted against the CRSP value-weighted index total return, for six horizons (1, 5, 10, 20, 40, and 60 trading days).

For this report specifically, the sample is narrowed and re-sorted as follows:

- **Event filter:** only events with a resolved day-0 trading session (`day0_status == 'ok'`), 261,537 of 268,606 candidate events (97.4%).
- **Surprise source:** restricted to `sue_source == 'analyst'` — i.e. events with a usable, point-in-time analyst consensus estimate — rather than pooling in the seasonal (year-over-year) fallback measure. The pipeline documentation notes these two measures are on different scales (sample sd 9.01 vs. 3.36); restricting to one source avoids that scale-mixing issue entirely for the headline result. This leaves **182,712 events** spanning 72 announcement quarters (1996–2013). The seasonal-sourced subsample is a natural robustness extension not included in this report — see Section 5.
- **Decile assignment:** within each announcement calendar quarter, analyst-sourced events are ranked by

sue_analyst (rounded to 8 decimals to avoid ordering exact ties by floating-point noise, matching the project's existing convention) and cut into ten equal-count groups. Ranking within quarter, not across the full sample, keeps the sort point-in-time-safe: a 1997 event's decile never depends on 2013 data.

Note on cell balance: because 11.6% of analyst-sourced events have a surprise of exactly 0.0000 (a large tied mass, documented in data/earnings/FINDINGS.md), deciles are not perfectly equal-sized where that tie mass falls — decile 4 has 22,423 events versus roughly 15,000–19,000 for most other deciles. This is a real feature of the data (heavy exact-zero-surprise mass, likely a mix of true zero surprises and coarse consensus rounding), not a bug in the sort, and does not affect the ranking's validity.

2. Inference: clustering by announcement quarter

The existing event-study output (data/results/event_study_stats.json) explicitly flags that its standard errors are plain i.i.d. ($sd/\sqrt{n}$) and therefore understate uncertainty, since earnings announcements cluster heavily within each fiscal quarter's reporting season. This report addresses that directly using the standard remedy in the PEAD literature: **Fama-MacBeth aggregation**. For each decile (and for the D10–D1 spread), the average return is computed separately within each of the 72 announcement quarters; the reported mean is the average of those 72 quarterly averages, and the standard error is the standard deviation of the 72 quarterly averages divided by $\sqrt{72}$. This is robust to arbitrary correlation among events within the same quarter (exactly the clustering concern raised in the source documentation), at the cost of basing inference on 72 quarterly observations rather than 182,712 firm-quarter observations — a conservative trade in favor of honest uncertainty.

3. Headline result: deciles drift as expected

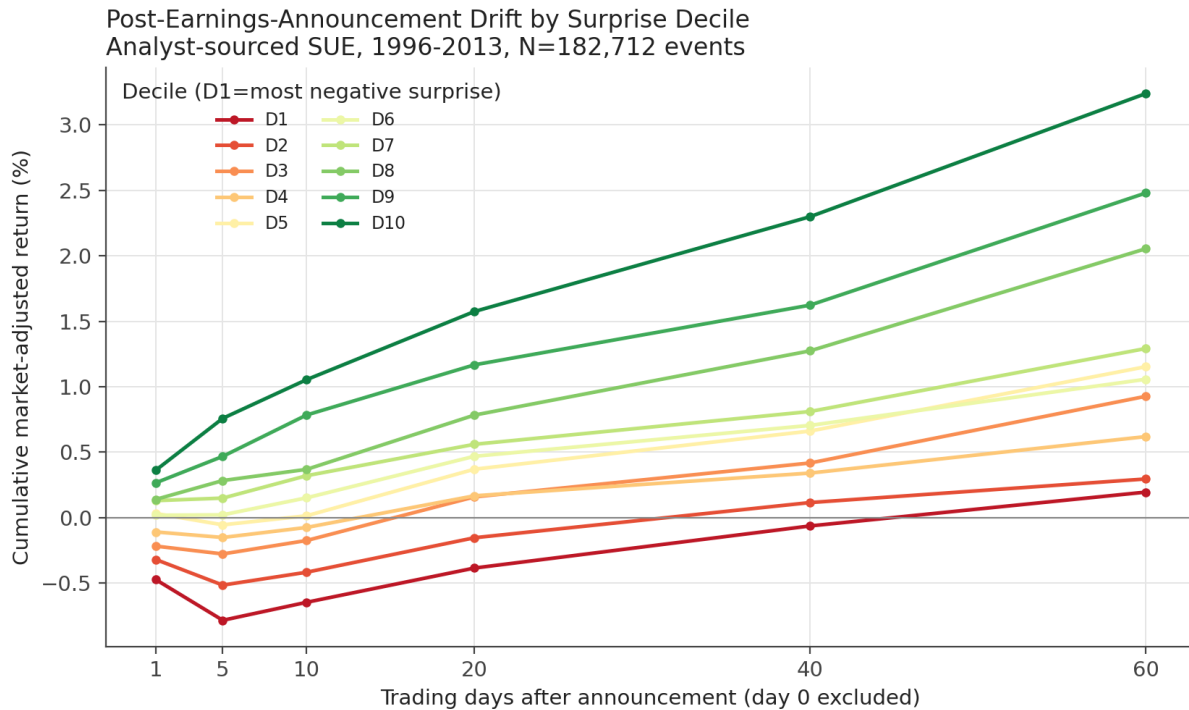


Figure 1. Cumulative market-adjusted return by SUE decile and horizon. D1 (most negative surprise, dark red) sits below zero at every early horizon and only crosses back above zero after 40+ trading days; D10 (most positive surprise, dark green) is positive from day 1 and rises steadily to +3.24% by day 60. The ordering across deciles is monotonic at every horizon from +10 days onward (a minor D4/D5 crossover appears only at the +1d and +5d marks, consistent with a small amount of short-horizon noise around the announcement itself, and resolves by +10d).

Decile	+1d	+5d	+10d	+20d	+40d	+60d
D1	-0.47%	-0.79%	-0.65%	-0.39%	-0.06%	+0.19%
D2	-0.32%	-0.52%	-0.42%	-0.15%	+0.11%	+0.29%
D3	-0.22%	-0.28%	-0.18%	+0.16%	+0.42%	+0.93%
D4	-0.11%	-0.15%	-0.08%	+0.17%	+0.34%	+0.62%
D5	+0.03%	-0.06%	+0.01%	+0.37%	+0.66%	+1.15%
D6	+0.02%	+0.02%	+0.15%	+0.47%	+0.70%	+1.06%
D7	+0.13%	+0.15%	+0.32%	+0.56%	+0.81%	+1.29%
D8	+0.14%	+0.28%	+0.37%	+0.78%	+1.27%	+2.05%
D9	+0.26%	+0.47%	+0.78%	+1.17%	+1.62%	+2.48%
D10	+0.36%	+0.76%	+1.05%	+1.57%	+2.30%	+3.24%

Table 1. Mean market-adjusted return (%) by decile and horizon (trading days after announcement). D1 = most negative surprise, D10 = most positive.

4. The D10 – D1 spread: the formal PEAD test

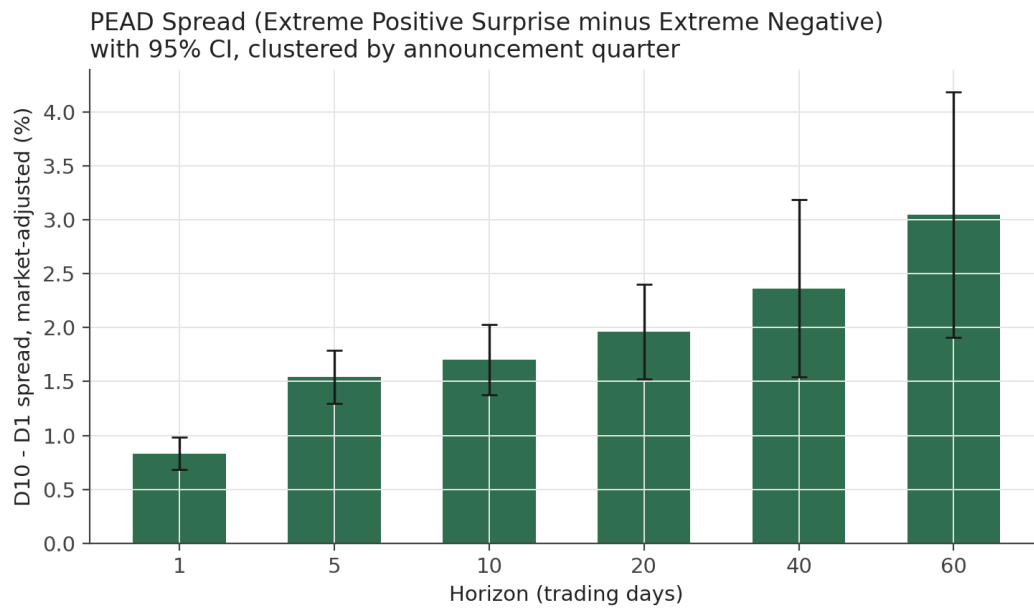


Figure 2. The spread between the top and bottom surprise deciles, with 95% confidence intervals from the quarter-clustered (Fama-MacBeth) standard error described in Section 2.

Horizon	D10-D1 spread	Clustered SE	t-stat	Quarters
+1d	+0.83%	0.08%	10.90	72
+5d	+1.54%	0.13%	12.23	72
+10d	+1.70%	0.17%	10.17	72
+20d	+1.96%	0.22%	8.74	72
+40d	+2.36%	0.42%	5.64	72
+60d	+3.05%	0.58%	5.25	72

Table 2. D10–D1 spread by horizon. All t-statistics exceed 5.2 even after quarter-clustering (versus the much larger, and overstated, t-statistics that plain i.i.d. SEs would produce) — the spread is not a small-sample or clustering artifact. The spread widens from +0.83% at 1 day to +3.05% at 60 days, consistent with genuine drift rather than a one-time reaction that happens to differ by decile.

4b. Robustness: is the drift just a size/value risk premium?

Section 3 showed that D1 (the most negative-surprise decile) does not fall monotonically in market-adjusted terms — after an initial drop through +5 days, it recovers and crosses back above zero by +60 days, even as D10 keeps climbing. One explanation raised for that pattern is that extreme negative-surprise firms tend to be smaller and more value/distressed than extreme positive-surprise firms, and small, high book-to-market stocks carry a higher unconditional expected return simply as compensation for risk — a Fama-French style size and value premium, not a market inefficiency. A single-factor market adjustment (Section 3's methodology) cannot separate that risk premium from a genuine mispricing correction. This section tests that directly.

Method. This session has no WRDS access and no verified source for the official published Fama-French factor series, so rather than pull an unvetted external file, this reconstructs the substance of the Fama-French size/value control from data already inside this project: book equity from Compustat ceq (funda.parquet has no txdtc, so $BE = ceq$ exactly, a standard simplification; firms with $ceq \leq 0$ are dropped, the usual convention since book-to-market is not meaningful for negative book equity), matched to each event with at least a 130-day reporting lag before the announcement to stay point-in-time-safe, divided by the event's own market cap to form book-to-market. Each event is placed into one of 15 cells (5 size quintiles $\times$ 3 book-to-market terciles, both computed within the same announcement quarter as the existing size-quintile convention). For every event, a **characteristic-adjusted return** is computed by subtracting the leave-one-out average market-adjusted return of every other event in the same quarter/size/book-to-market cell — so each event is benchmarked against economically similar firms reporting around the same time, not just the overall market.

Coverage: 175,135 of 182,712 events (95.9%) matched a point-in-time book-equity figure; 175,016 (95.8%) had both a size quintile and a book-to-market tercile and could be characteristic-adjusted. The decile sort itself is unchanged from Section 3 — only the outcome variable (raw market-adjusted return vs. characteristic-adjusted return) differs.

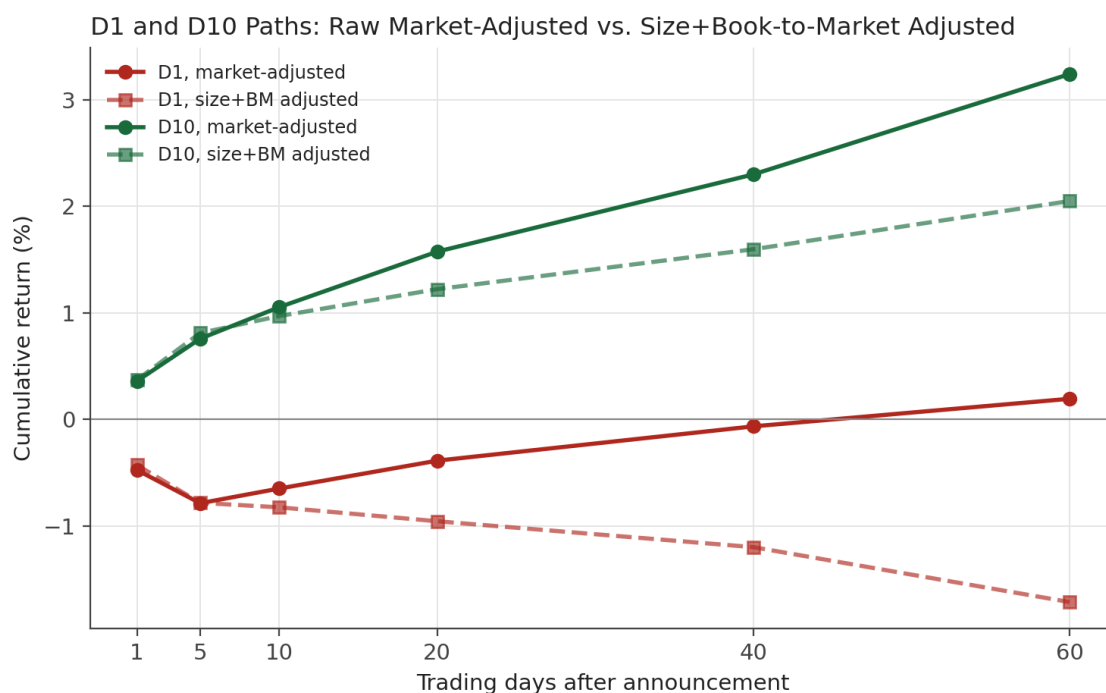


Figure 6. D1 and D10 cumulative return paths under the two adjustments. Once size and book-to-market are controlled for (dashed lines), D1 no longer recovers — it declines monotonically to -1.71% by +60 days, exactly the pattern the underreaction story predicts for bad news. D10's path is barely changed by the adjustment. This indicates that a meaningful share of D1's apparent "recovery" in the raw market-adjusted numbers was in fact a size/value risk premium riding on top of the drift, not evidence against PEAD.

Horizon	Raw D10-D1	Size+BM-adj. D10-D1	Adj. t-stat	Quarters
+1d	+0.83%	+0.80%	11.41	72
+5d	+1.54%	+1.59%	14.21	72
+10d	+1.70%	+1.79%	12.30	72
+20d	+1.96%	+2.18%	12.09	72
+40d	+2.36%	+2.79%	9.68	72
+60d	+3.05%	+3.76%	9.25	72

Table 3. D10–D1 spread, raw market-adjusted vs. size+book-to-market adjusted. The adjusted spread is, if anything, larger than the raw spread at every horizon (+3.76% vs. +3.05% at 60 days) and remains highly significant (t-stats 9.25–14.21) after the same quarter-clustering as Table 2. The drift is not explained away by size or value risk exposure.

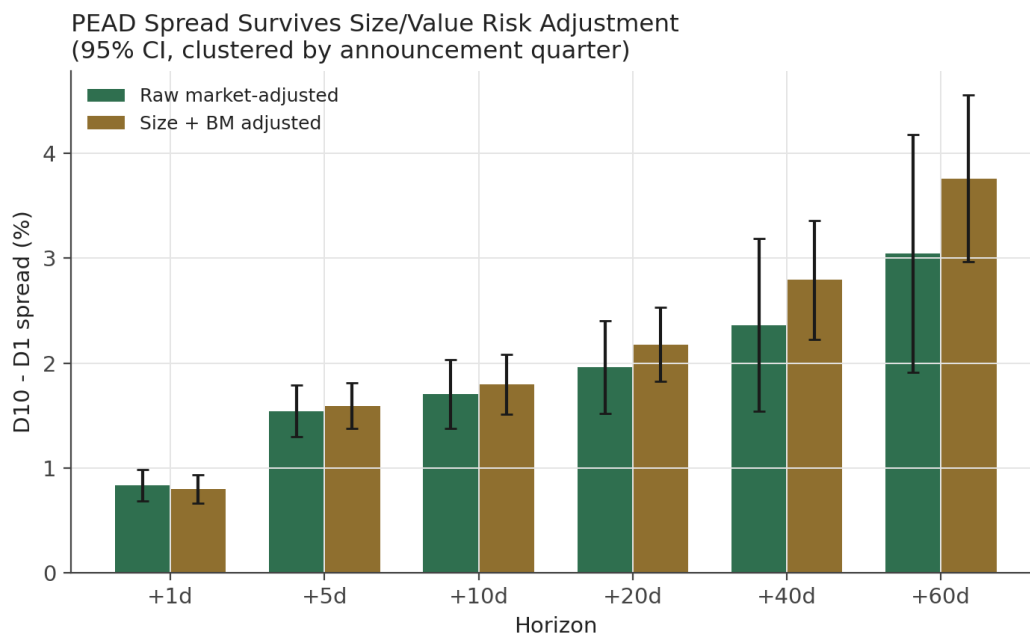


Figure 7. Side-by-side comparison of the raw and characteristic-adjusted D10–D1 spread across all six horizons, with 95% confidence intervals.

Reading this result honestly: this is a characteristic (cell-mean) adjustment against other events in this same restricted universe, not a regression against externally-published, whole-market Fama-French factors — see the note on universe scope in Section 1 and the caveat in Section 6. It controls for size and value tilts within this project's own sample well, but it would not catch a risk exposure this universe is itself skewed on relative to the broader market (e.g. if the OptionMetrics-linked universe as a whole is more mega-cap-tilted than the total CRSP universe). A regression against the officially published Fama-French factors remains a natural follow-up once either WRDS access or a verified factor-file source is available in this environment.

5. Does the effect vary across industries, firm sizes, and eras?

A natural follow-up question is whether PEAD is a uniform phenomenon or concentrated in particular kinds of companies. This section reuses the classification features already built earlier in the project (data/results/event_classification_features.parquet: Fama-French 12-industry sector, firm-size quintile at announcement, and a 5-bin era) and asks, for each grouping variable, whether the D10–D1 spread at the +60d horizon differs across groups — using the same Fama-MacBeth clustering as above, plus a one-way ANOVA across each group's 72 (or fewer, where a group lacks quarters) quarterly spread observations as a formal significance test of whether the groups can be pooled.

5a. Industry sector — no significant difference; safe to pool

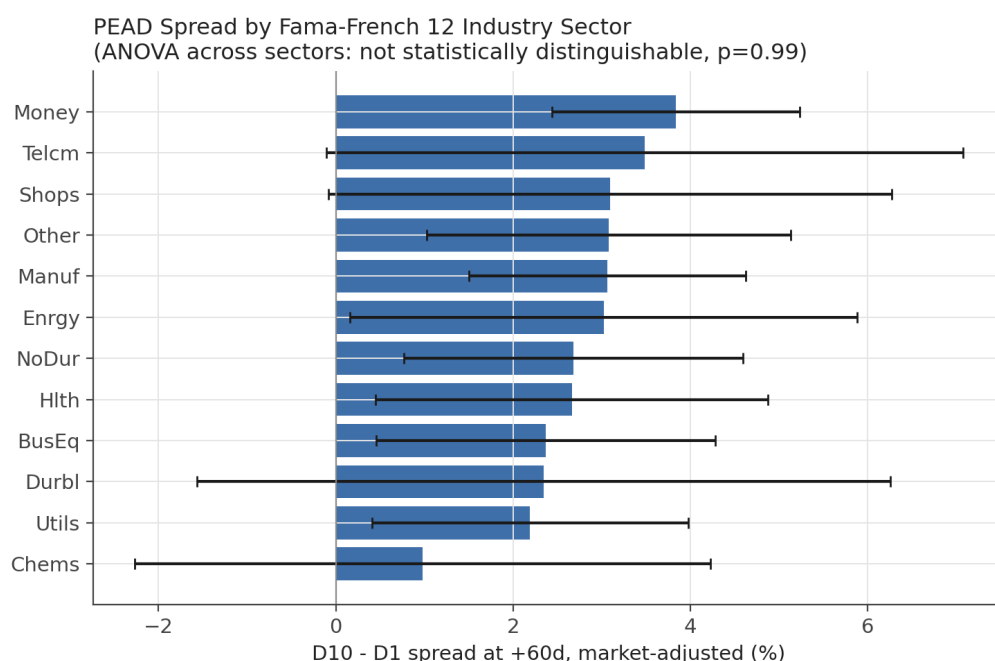


Figure 3. D10–D1 spread by Fama-French 12 sector, 95% CI. One-way ANOVA across the twelve sectors: $F=0.30$, $p=0.986$ — the null of equal average spread across sectors cannot be rejected. Every sector's point estimate falls in the 1.0%–3.8% range and every confidence interval overlaps every other sector's.

Implication: this is a genuinely useful grouping result — the drift effect is economically similar in magnitude across Financials (Money), Business Equipment, Healthcare, Manufacturing, Energy, and every other FF12 sector tested. A trading or research application does not need sector-specific decile cutoffs or sector-specific position sizing; one pooled model across sectors is statistically justified by this data.

5b. Firm size — a strong, monotonic gradient; should NOT be pooled

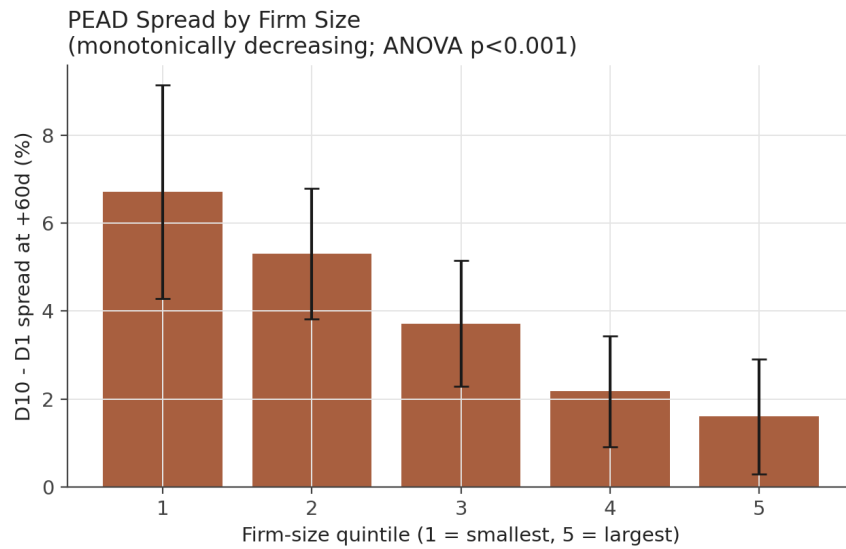


Figure 4. D10–D1 spread by firm-size quintile at announcement (1=smallest, 5=largest). One-way ANOVA: $F=6.50$, $p<0.0001$. The spread falls monotonically from 6.71% (smallest quintile) to 1.60% (largest quintile) — more than a 4x difference in effect size.

Implication: unlike sector, size is not safe to pool over. This matches the standard explanation in the literature: PEAD is a limits-to-arbitrage phenomenon — small, illiquid, less-covered stocks are harder and more costly for sophisticated investors to trade against, so the mispricing persists longer and larger. A strategy or further study built on this data should condition on size (e.g. report results by size tercile/quintile, or explicitly test whether the effect survives after transaction-cost-realistic position sizing in the smallest quintile) rather than reporting one pooled number.

5c. Era — suggestive decline, not statistically conclusive

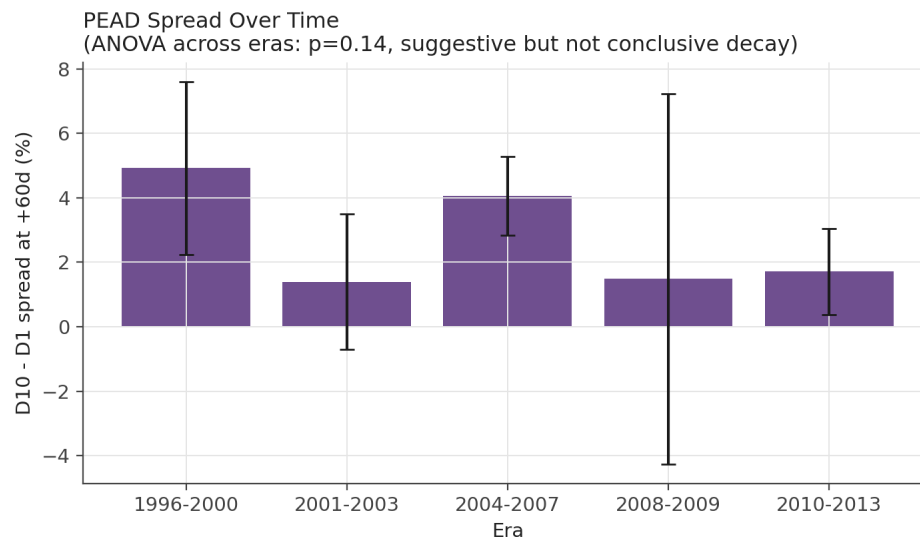


Figure 5. D10–D1 spread by 5-bin era. One-way ANOVA: $F=1.81$, $p=0.138$ — not significant at conventional levels, though the point estimates are suggestive: the spread is largest in 1996–2000 (4.92%) and 2004–2007 (4.06%), and smallest in 2001–2003, 2008–2009, and 2010–2013 (1.4%–1.7%).

Implication: this is the one place this report is deliberately cautious rather than declarative. A pattern consistent with the literature's own narrative — PEAD shrinking over time as more capital specifically targets it — is visible in the point estimates, but with only 5 era buckets and non-trivial within-era volatility (note the wide 2008-2009 CI, driven by only 8 quarters and the 2008 crisis's own volatility), the ANOVA does not clear a conventional significance bar. This should be read as a hypothesis for further work (e.g. a rolling-window version of Table 2 rather than 5 fixed bins), not a confirmed finding.

6. From signal to strategy: three portfolio constructions

Sections 3-5 establish that the surprise-decile signal is real, survives a risk adjustment, and is broadly consistent across industries. This section turns that signal into an actual traded portfolio and asks what it would have earned, net of transaction costs and realistic position-size limits. Three different ways of turning the ranking into positions are built and compared side by side, since they make different tradeoffs between simplicity, using the full signal, and controlling for the size effect found in Section 5b.

Holding period, decided from the data rather than assumed. Instead of guessing how long to hold a position, the size+book-to-market-adjusted return (Section 4b's cleaner measure) was extended out to 80, 100, 120, 150, and 180 trading days (beyond the 60-day ceiling used everywhere above), and each decile's *marginal* return, the extra return earned specifically between two adjacent checkpoints, was tracked to find where it flattens toward zero.

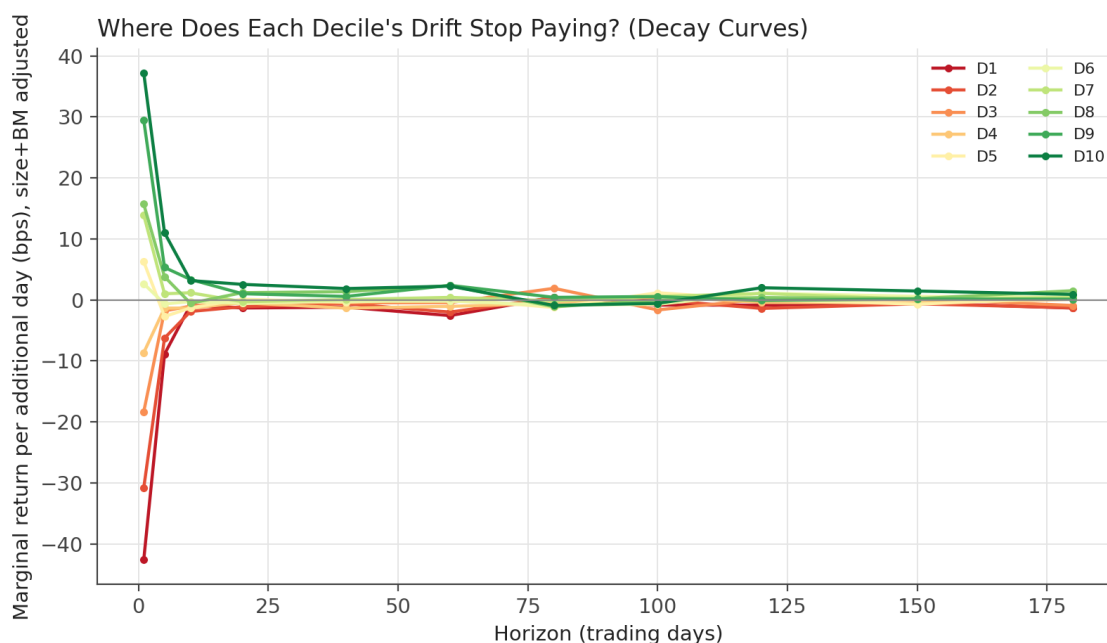


Figure 11. Marginal return per additional trading day of holding, by decile, size+book-to-market adjusted. Every decile's marginal contribution is large and directional in the first 5-10 days, then settles toward a small, noisy band close to zero by roughly day 20-40. The extreme deciles (D1, D2, D9, D10) keep contributing a small but consistently signed amount out to 80 days; the near-median deciles (D5, D6) flatten out almost immediately, consistent with their weak, often statistically insignificant drift throughout this report.

From this, three holding-period buckets are used everywhere below: **extreme** deciles (1, 2, 9, 10) hold 80 trading days; **shoulder** deciles (3, 4, 7, 8) hold 40 trading days; **near-median** deciles (5, 6) hold 20 trading days, a short duration reflecting that there is little measured edge there to wait for. This is a deliberate simplification of a noisier per-decile decay estimate (some individual deciles' exact flattening point moves around from measurement noise, especially the weak-signal middle ones) into three defensible, coarser bands rather than chasing every noisy inflection point.

Strategy 1: extreme decile long/short. Long every D10 event, short every D1 event, weight ± 1 , nothing else traded. This is the closest to the original naive framing and to the report's own headline test.

Strategy 2: rank-weighted. Every event, across all ten deciles, gets a position weight of $2 \times (\text{percentile rank} - 0.5)$, continuous from -1 (worst surprise that quarter) to $+1$ (best), close to zero near the median. This uses the entire cross-section the report has already validated (every decile drifts in the expected direction, not just the extremes) instead of discarding 80% of events at a hard cutoff. Each event's holding period still comes from its own decile's bucket above.

Strategy 3: extreme decile, size-balanced. Same D10/D1 selection as Strategy 1, but each leg's weight is redistributed across firm-size quintiles within each announcement quarter so every size quintile contributes an equal total weight to that leg. This directly answers the Section 5b concern: without this, the long and short legs could differ in average firm size purely by chance in a given quarter, secretly turning part of the "PEAD return" into an unrelated small-cap-short/large-cap-long bet.

Position sizing and costs. Every position is sized at a *fixed* dollar notional (weight $\times$ \$10,000 per unit of full weight), never as a percentage of the strategy's own live, compounding NAV — sizing off a growing NAV is what caused an earlier, now-discarded prototype's balance to inflate to an impossible \$1.05 trillion from a \$1M start, and fixed-notional sizing is the direct, structural fix, not a parameter tweak. A position's notional is capped at 1% of the stock's own 21-day average dollar trading volume (already computed in this project's classification features) so the backtest never assumes it could trade an illiquid name at a size the real market couldn't absorb; where this cap binds, the position is scaled down; it never happens for more than 18% of positions in any strategy tested here. Transaction costs are charged on both entry and exit, tiered by liquidity quintile from 5 basis points (most liquid) to 25 basis points (least liquid) one-way — a reasonable, disclosed assumption, not one calibrated to any specific broker or venue, since no such data exists in this project.

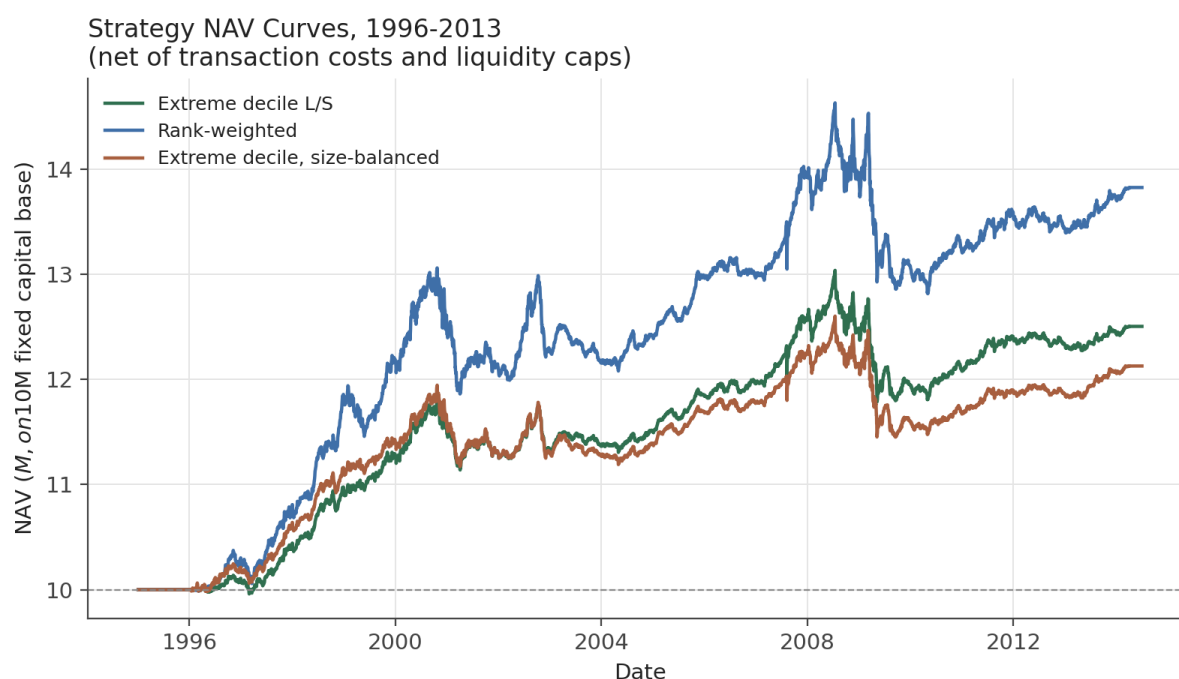


Figure 8. Net-asset-value curves for all three strategies, starting from a fixed \$10M capital base, net of transaction costs and liquidity caps. All three are profitable over the full window; the rank-weighted strategy (which trades the entire cross-section rather than only the extreme 20%) compounds the most.

Metric	Extreme L/S	Rank-weighted	Size-balanced
Annualized return	1.15%	1.68%	1.00%
Annualized volatility	2.31%	3.61%	2.23%
Sharpe ratio	0.56	0.54	0.49
Max drawdown	-9.60%	-12.40%	-9.13%
Avg. leverage (gross/capital)	0.51x	1.07x	0.47x
Avg. concurrent positions	558	1,843	558
Annual turnover (x capital)	3.2x	8.4x	3.0x
Total transaction costs	\$625,938	\$1,640,524	\$580,627
% positions liquidity-capped	18.0%	8.3%	18.2%

Table 4. Strategy comparison, 1996-2013, net of transaction costs. All three strategies show a positive Sharpe ratio after realistic costs and liquidity limits; the rank-weighted strategy has the highest return and Sharpe but also the highest turnover, leverage, and drawdown, an intuitive tradeoff, since it trades far more names (173,445 positions vs. 34,225) and carries a position, even a small one, on almost every reporting event rather than only the extreme 20%.

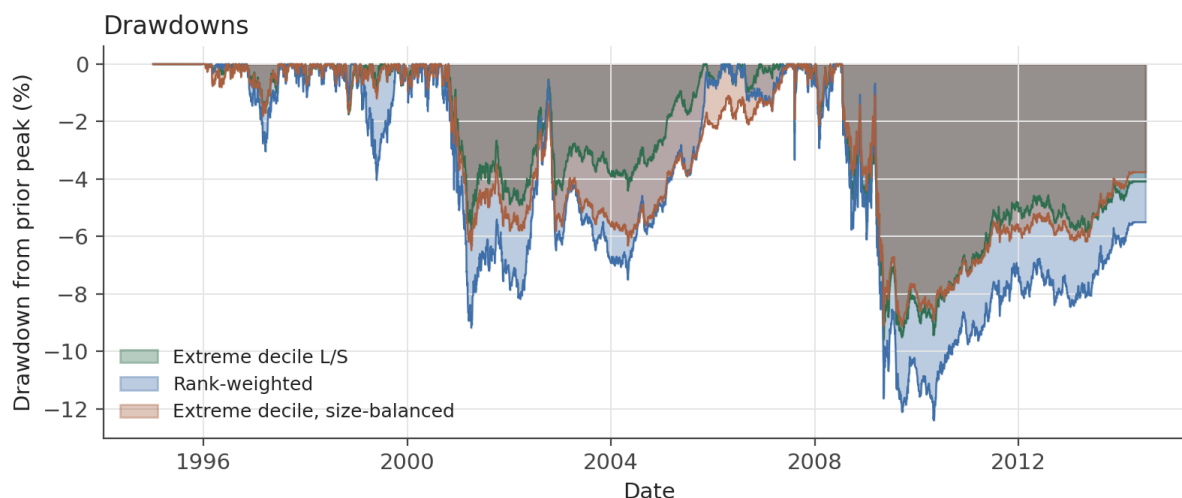


Figure 9. Drawdown from the prior peak, all three strategies. The largest drawdowns for every strategy cluster around the 2008-2009 financial crisis, unsurprising given the market-adjusted (not fully risk-adjusted) return construction and the general spike in volatility and correlation across nearly all equities during that period.

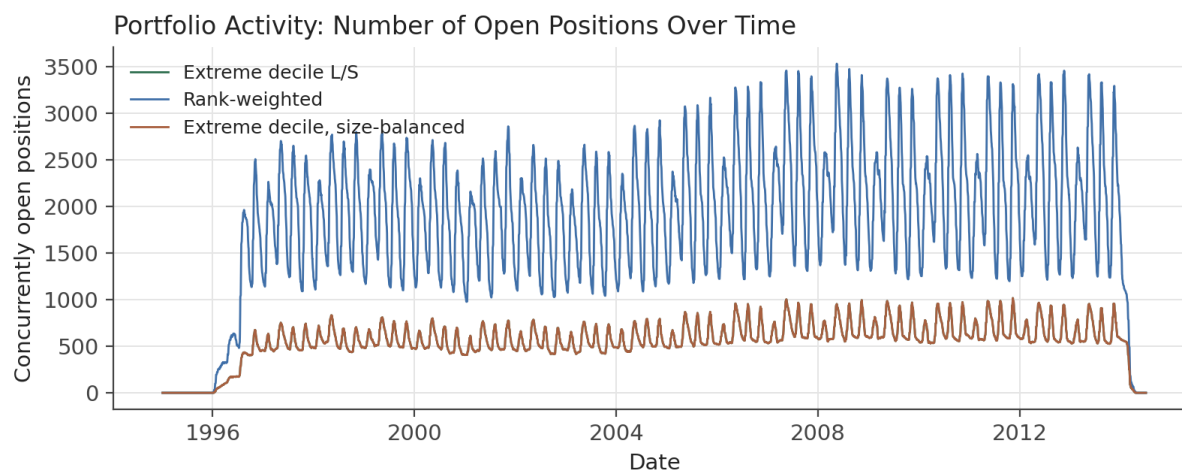


Figure 10. Number of concurrently open positions over time. The rank-weighted strategy runs roughly 3.3x as many simultaneous positions as the two extreme-decile strategies (which are identical in position count, differing only in how weight is distributed within each leg), directly because it trades every decile instead of only the top and bottom.

Reading these results honestly. These are backtested, not tradeable-today, numbers, and several simplifications are disclosed together in Section 7 below: position sizing is on a fixed capital base rather than true compounding, transaction cost assumptions are not calibrated to real execution data, and short-side borrow cost and availability (a real constraint specifically for the small, less-liquid names that dominate D1) are not modeled at all. The Sharpe ratios here (0.49-0.56) are modest by design, choices here consistently favored realism (liquidity caps, costs, conservative fixed sizing) over showing the largest possible number.

7. Caveats carried over from the underlying data pipeline

This report inherits several caveats already disclosed in the project's own data documentation, restated here because they bear directly on how to read the results above:

- **Survivorship in longer horizons.** `data/events/FINDINGS.md` documents that return coverage shrinks from 96.8% at +1d to 95.7% at +60d, almost entirely because a security's own price panel ends (predominantly delisting) before the target date. Longer-horizon statistics in this report implicitly condition on surviving that long — attrited events have a significantly more negative average surprise than retained ones ($t=-4.83$ per the source documentation), so this attrition is not random and could modestly understate the true magnitude of drift for the worst-surprise decile at long horizons specifically.
- **Reaction/drift conflation at the shortest horizon.** Daily-frequency data cannot fully separate an announcement's own price reaction from its first day of drift when the announcement lands during market hours; some of the "reaction" at +1d may already be drift, or vice versa.
- **No transaction costs in Sections 3-5.** The headline decile and subsample returns (Sections 3-5) are gross, unlevered, and do not account for bid-ask spreads, market impact, or borrowing costs for the short leg — a material omission specifically for the smallest-size quintile in Section 5b, where trading costs are typically highest and the raw spread is largest. Section 6's backtest does apply costs and liquidity caps, but see its own caveats below.
- **Sector/size/era classification coverage.** The classification features file matched 100% of the analyst-sourced sample used here, but carries its own upstream flags (e.g. SIC-code fallbacks) documented separately in the results pipeline; not re-audited for this report.
- **Quarter-level clustering only.** Fama-MacBeth clustering by announcement quarter addresses earnings-season clustering but not, for example, clustering by industry-day if multiple same-industry firms report on the same date within a quarter; a firm-and-time double-clustered standard error (e.g. via a panel regression) would be a stronger — and more complex — alternative for future work.
- **Universe scope.** Restated from Section 1: every result in this report, including the Section 4b risk adjustment, is confined to the OptionMetrics-linked universe (~8,100 permnos) that this project's WRDS pull already used. It is a broad cross-section but excludes any stock that never had listed options, and cannot be expanded without a new WRDS pull this session cannot perform.
- **The size/BM adjustment is characteristic-based, not a factor regression against official Fama-French series.** Section 4b benchmarks each event against other events in the same universe, same quarter, same size/book-to-market cell — it controls for size and value tilts within this sample, but would not catch a risk exposure the OptionMetrics universe is itself systematically skewed on relative to the full market. Book equity also uses `ceq` alone (no deferred-tax adjustment, unavailable in this pull) and a 130-day fixed reporting lag rather than the exact July-June Fama-French timing convention.
- **Backtest: fixed-notional, not compounding.** Section 6 deliberately sizes every position off a fixed dollar base rather than the strategy's own live NAV, to avoid the runaway compounding bug found in an earlier prototype. A real fund would periodically resize its base off a trailing NAV; this backtest reports return on a fixed capital base instead, which understates what true reinvestment would have compounded to in a genuinely profitable period, and overstates it in a genuinely unprofitable one.
- **Backtest: transaction costs and liquidity caps are disclosed assumptions, not calibrated figures.** The 5-25bps one-way cost schedule and the 1%-of-ADV position cap in Section 6 are reasonable, conservative-leaning choices, not fitted to any real broker, venue, or historical execution data, since none exists in this project.
- **Backtest: no borrow cost or short-availability modeling.** Every strategy in Section 6 that shorts D1 assumes the short is executable at the modeled cost with no separate borrow fee and no risk the shares simply aren't available to borrow — a real constraint, worse for exactly the small, illiquid names that dominate the negative-surprise decile, and not modeled here at all.
- **Backtest: one historical realization, not walk-forward validated.** Section 6 reports one full-sample backtest over 1996-2013; none of this report's design choices (holding-period buckets, cost tiers, liquidity cap) were tuned on a held-out period and then validated out of sample, so there is no guarantee they wouldn't have been chosen

differently, and performed differently, if fit only on an earlier subperiod and tested on a later one.

- **Data ends in 2013, and PEAD is a well-known, heavily studied anomaly.** A live strategy today would need current data revalidation; the era analysis in Section 5c already found weaker point estimates in the most recent (2010-2013) years of this sample, consistent with the literature's own narrative of the effect shrinking as more capital targets it.

8. Suggested next steps

- Extend Section 5's sector/size/era cuts to interaction terms (e.g. size x era) once cell sizes allow it — several era buckets are already thin (8–12 quarters) and would need care before slicing further.
- Add the seasonal-sourced (`sue_source == 'seasonal'`) subsample as an explicit robustness cut on its own scale, rather than folding it into the pooled `sue_rank_pct` the underlying pipeline already computes.
- Build a day-by-day (rather than fixed-horizon) cumulative abnormal return curve using the daily CRSP panel directly, to see the drift's exact shape rather than six snapshot points — this report used the six pre-computed horizons already in the event table for efficiency, but a full daily curve would make the "no announcement-day jump, then steady drift" story even more visually direct.
- Replace Section 4b's within-universe characteristic adjustment with a regression against the officially published Fama-French factor series (and consider adding momentum/profitability/ investment factors for a full 5-factor check) once this environment has WRDS access or a verified factor-file source — see Section 7's caveat on this.
- Expand the universe past the OptionMetrics-linked ~8,100 permos if the full CRSP/IBES universe is wanted — requires a new WRDS pull outside this session (see Section 1).
- Walk-forward validate Section 6's strategies: fit holding-period buckets and cost/liquidity parameters on an early subperiod only, then test unchanged on a later held-out period, rather than the single full-sample backtest reported here.
- Resize position notional off a trailing (not same-day) NAV so the backtest reflects true compounding, replacing the fixed-capital-base approximation used in Section 6.
- Model short-side borrow cost and availability explicitly, particularly for D1, where the smallest and least liquid names are concentrated and real-world shorting frictions are worst.
- Calibrate the transaction cost and liquidity-cap assumptions against real execution data (actual spreads, actual market impact) if or when such data becomes available to this project, replacing the disclosed-but-uncalibrated schedule used here.
- The empty `data/event_options/` and `data/raw_options/` folders in this project suggest an options-based version of this strategy was anticipated (e.g. buying calls on D10 events, puts on D1 events) -- that would sidestep the short-borrow problem entirely and add convexity, and is a natural next build once options data is pulled.